3 A ball is thrown vertically upwards. It reaches its maximum height of motion and comes back to the original point where it was launched.

Given that air resistance to the motion of the ball is not negligible, which of the following statements about the ball is correct?

A The ball is accelerating at $-9.81\ \mathrm{m\ s^{-2}}$ throughout its motion.

B The ball is decelerating at a decreasing rate when it is moving upwards.

C The ball is decelerating at an increasing rate when it is moving upwards.

D The ball is accelerating at an increasing rate when it is moving downwards.

4 A ball of weight W moves along a smooth horizontal surface until it falls off the edge at time